MILITARY SPECIFICATION

VALVE, STEAM PRESSURE REDUCING,

QUIET DESIGN

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers quiet design, self-contained steam pressure reducing valves for submarines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-R-17131 - Rods, Welding, Surfacing.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B1.12 - Class 5 Interference - Fit Thread.
B16.5 - Steel Pipe Flanges and Flanged Fittings.
B18 - Bolts and Nuts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)
A105-73 - Forgings, Carbon Steel for Piping Components.
A193-74 - Alloy-Steel and Stainless Steel Bolting Materials for High Temperature Service.
A194-73 - Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service.
A216-74 - Carbon, Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A306-66 - Carbon Steel Bars Subject To Mechanical Property Requirements.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philladelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Valves furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Valve description. Valves shall be a self-contained, internal pilot operated type. Downstream pressure feedback shall be sensed by a spring-loaded diaphragm to position a pilot valve which utilizes inlet steam pressure to position the main valve plug via an operating piston.

3.3 Materials. Materials shall be as specified in table I. Materials for parts not listed in table I shall be corrosion-resistant, suitable for the intended pressures and temperatures, and selected to prevent galling, seizing, or excessive wear on operating parts. Clearances shall prevent interferences due to the thermal expansion.

Table I - List of materials.

Name of part	Materials
Body, bonnet, and bottom cover	ASTM A105
	ASTM A216, grade WCB
	ASTM A306, grade 60
Internal trim (main valve and pilot), plug, seat ring, guide posts and bushings	Stellite (see note) Hardened 400 series stainless steel
Cylinder liner and piston	400 series stainless steel with surface hardness of 500 Brinell minimum (440°C shall be used only when approved)
Gaskets	MIL-G-21032, class B
Diaphragm	Nickel-copper alloy
	300 series stainless steel
	Nickel-chrome alloy
Springs	Inconel X-750
	A-286
	Inconel 600
	Tungsten tool steel
Bolting	ASTM A193, grade B7
	ASTM A194, grade 2H

NOTE: Where stellite is specified, it may consist of either wrought stellite 6B, cast stellite 6, or an inlay of stellite (not less than 3/32 inch thickness for main seat and disc surfaces). Where inlays are used, welding rods shall be in accordance with type MIL-RCoCr-A of MIL-R-17131.

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3.4 Design and construction.

3.4.1 Design concept. Valves will be operated, maintained, and repaired onboard Naval ships and shall emphasize simplicity, maintainability, ruggedness, and reliability. Design shall permit access for adjustment and repair when working from either side of the valve and without requiring removal of the valve body from the line.

3.4.2 Maintainability. Internal parts shall permit easy disassembly and reassembly with standard tools and shall prevent, as far as practical, the incorrect reassembly of parts. Positioning and alinement of all parts in assembly shall employ positive means so that correct reassembly is repeatedly assured. In no case shall parts for a given valve be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.4.2.1 Valve design shall permit accomplishment of the following maintenance actions within the time limits specified:

Action	Time allowed
Renew pilot valve assembly trim	1/2 hour
Renew main valve trim	3/4 hour
Remove, clean, and replace quiet flow regulating element	2-1/2 hours

3.4.3 Interchangeability. Valve design shall permit interchangeability without individual modification of like parts between all valves. Each part shall have part number identity and shall be replaceable from stock or the manufacturer on a non-selective and random basis. Parts having the same manufacturer's part number shall be directly interchangeable with each other with respect to installation (physical) and performance (function). Physically interchangeable assemblies, components, and parts are those which are capable of being readily installed, removed, or replaced without alteration, misalinement, or damage to parts being installed or to adjoining parts. No fabrication operations such as cutting, filing, drilling, reaming, hammering, bending, prying, or forcing shall be required. This is not intended to preclude the use of special tools, fixtures, and other shop aids during original assembly of the parts into the article. However, any special tools required for maintenance shall be supplied with the valve. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.4.4 Springs. Springs shall not be fully compressed during any normal operation or adjustment of the valve. The working stress shall be such that relaxation shall not exceed 5 percent over a 1000 hour period at the nominal operating temperature. Spring ends shall be squared and ground.

3.4.5 Threads. Threads shall conform to Handbook H28. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used. ANSI B-18 hex-head standards shall be used.

3.4.6 Pressure-temperature rating. Pressure-temperature rating shall be in accordance with 600 pound series in accordance with ANSI B16.5.

3.4.7 End preparation. End connections shall be in-line 2-1/2 inch iron pipe size (ips), 600 pound series, raised-face flanges in accordance with ANSI B16.5. Face-to-face distance between the raised-faces shall be 13 inches (± 1/16 inch). Valves shall be capable of being installed, operated, adjusted, and maintained within the envelope defined by figure 1. Inlet and outlet bore diameter shall be 2.320 ± 0.010 inches.

3.4.8 Bonnet and bottom cover joints. Bonnet and bottom cover (where applicable) shall be flanged for attachment to the body. Joints shall be secured by either of the following:

(a) Through-bolts threaded the entire length and fitted with a nut on each end. Threads on both bolts and nuts shall be class 2 fit in accordance with Handbook H28.
(b) Tap-end studs with interference fit at the tap end and a class 2 fit at the nut end. Interference fit shall be in accordance with ANSI B1.12.

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Bonnet and bottom cover shall be located by body guiding (i.e. a close tolerance fit between machined diameters on the body, bonnet, and bottom cover) rather than depending on studs or bolts for location. Spiral wound gaskets shall be fully retained and the joints shall have metal-to-metal take-up to provide controlled compression of the gaskets. To assure easy gasket removal, not more than two gasket-retaining faces for each gasket shall be formed on a single part. Joint design shall assure parallel alinement of the guide bushings. Sufficient bolting area shall be provided to maintain metal-to-metal make-up over at least a 3 year period. Bearing surface of nuts and their respective surfaces on the valve shall be finished machined.

3.4.9 Body construction. Valve bodies shall be machined from a one-piece casting or forging and be a basic globe configuration with in-line inlet and outlet ports. All steam lines, including the impulse line, shall be internally ported in the body and bonnet.

3.4.10 Internal trim.

3.4.10.1 Main valve. The main valve internal trim shall be of cage construction. The quiet-flow regulating element and the seat ring shall be clamped in position by a cage which is either part of, or held in position by, the bonnet or bottom cover flange. This shall permit rapid replacement of all internal trim by avoiding the use of any threads within the valve cavity (such as seat-ring threads). Design shall assure that the cage remains under sufficient compression at all times to avoid any movement or looseness and the quiet-flow regulating element shall be confined and compressed on both top and bottom by this cage construction.

3.4.10.1.1 Quiet-flow regulating element. The element design employed to achieve a low-noise pressure loss shall avoid excessively small flow passages or any other feature jeopardizing operational reliability. It shall be brazed or otherwise fabricated into a one-piece configuration, capable of withstanding installation and operational stresses. Internal passages shall be such that a 0.033 inch diameter perforated metal screen type strainer installed upstream will protect against clogging. Element cleaning shall not be required more often than annually, and replacement due to blockage, erosion, wear, etc., more often than once each 5 years. The element shall be capable of being quickly cleaned by backflush or with ultrasonic equipment. Special fixtures necessary for cleaning the element shall be supplied with each valve. The element structure shall be of a solid or continuous nature, not susceptible to media migration.

3.4.10.1.2 Operating piston. The operating piston shall be separate from the main valve and fitted with one or more piston rings. The design shall prevent water build up on the piston. The piston shall operate within a separate hardened-steel cylinder liner located in the valve body so that removal of the valve bonnet provides access to the top of the piston assembly. The cylinder liner shall be held in place by way of the bonnet bolting or shall be permanently fabricated into the body. The requirement to locate the cylinder lever in the body may be waived where it is shown that an alternate location provides a satisfactory maintenance configuration.

3.4.10.2 Pilot valve. A quiet-flow throttling element, in accordance with 3.4.10.1.1, shall be employed in the pilot system, as necessary, to meet the overall noise criteria of 3.6.5. The pilot valve seat shall be either directly threaded into the bonnet or retained by a cage construction. In either case, the seat shall be gasket sealed. Pilot valve and diaphragm chambers shall be self-draining. The pilot valve shall be single seated with integral stem. The valve shall be controlled by a spring referenced metal diaphragm and shall open against high-pressure. A return spring shall keep the pilot valve in contact with the diaphragm at all times. The diaphragm shall not travel through center during any phase of operation. Edges contacting the diaphragm shall be rounded to prevent wear and damage.

3.5 Service and ambient conditions. Service and ambient conditions are as follows:

(a) Fluid - Saturated steam; 0 to 1 percent moisture.
(b) Inlet pressures - 250 to 740 pounds per square inch gage (lb/in^2g).
(c) Design pressure - 900 lb/in^2g (strength only).
(d) Temperature - Saturation temperature.
(e) Flow demand - Minimum: tight shut-off as defined in 3.6.3.
(f) Range of sustained flow demand - 500 to 10,000 pound per hour (lb/hr).
(g) Rate of flow demand variation - 800 lb/hr/sec maximum.
(h) Rate of inlet pressure variation - 30 lb/in^2g/sec maximum.
(i) Shock - As defined by grade A, class I of MIL-S-901.
(j) Vibration - As defined by type I of MIL-STD-167-1.

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3.6 Performance requirements.

3.6.1 Accuracy of regulation. Accuracy of regulation, defined as the maximum permissible variation in downstream pressure which can occur with the valve set anywhere within the required range of adjustment and then subjected to any number of combinations of service and ambient conditions defined in 3.5, shall be 20 lb/in^{2}g.

3.6.2 Capacity. Valve is required to shut-off tightly as defined in 3.6.3. The valve will not be subjected to sustained flow demands between shut-off and 500 lb/hr. Between flow rates of 500 lb/hr and 10,000 lb/hr, at any set point within the range defined in 3.6.4, and at any inlet pressure and transient condition defined in 3.5, the valve shall operate smoothly without hunting, chattering, exceeding the noise criteria specified in 3.6.5, or violating the accuracy of regulation specified in 3.6.1.

3.6.3 Seat tightness. The valve shall meet the following seat tightness requirements: With a dead-end downstream volume not exceeding the volume represented by 100 diameters of downstream pipe, any steam leakage from the inlet to the outlet of the valve shall be limited below a value which will cause a discharge pressure buildup of more than 10 lb/in^{2}g in a 1 hour period.

3.6.4 Range of adjustment. The valve shall be adjustable over a range of delivered pressures of 130 to 200 lb/in^{2}g.

3.6.5 Noise. The valve shall meet the structureborne noise requirements as defined in MIL-STD-740 for type 2 resiliently mounted equipment as measured on the valve body and connecting flanges both parallel and perpendicular to the direction of flow for any combination of operating conditions within the specified limits.

3.6.6 Mechanical shock and vibration. Valves shall be designed to meet the mechanical shock requirements defined by grade A, class I of MIL-S-901 and the environmental vibration requirements defined by type I of MIL-STD-167-1.

3.7 Marking.

3.7.1 Body markings. The manufacturer's name or trademark and the body material composition shall be cast or forged integral with the valve body. The size, rating, and a flow arrow shall be cast or forged integral with the valve body or die stamped on raised metal pads (1/8 inch added wall thickness minimum).

3.7.2 Identification plates. Each valve shall have an identification plate permanently attached to an exposed position on the valve that will not be covered by insulation. The identification plate shall be made of corrosion-resistant material and shall contain the following information or a space therefor:

(a) Manufacturer's name and address.
(b) MIL-V-24517.
(c) ANSI 600 pound rated.
(d) Adjustable set pressure range.
(e) Manufacturer's model and part number or identification.
(f) Manufacturer's drawing numbers (assembly and CD).
(g) Technical manual number.
(h) 9-digit CID/APL number.

3.8 Technical data. The supplier shall prepare engineering drawings and a technical manual in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.8.1 Drawings. In addition to the drawing content required by the data ordering document, the following unique technical features shall be included:

(a) Calculated average fluid velocity through various portions of the valve. These shall include the inlet and outlet ports and, at minimum, enough sections within the valve to cover any velocity change in excess of twenty-five percent.
(b) Construction details of the "quiet" throttling element, including: (1) data on maximum particulate size that can be passed without clogging, (2) brief discussion or rationale on the mechanics by which pressure breakdown is achieved without generating the noise

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level normally associated with orifical throttling, (3) method of fabrication: configuration and process of fabrication shall be described in sufficient detail to illustrate that adequate structural strength and freedom from flow passage obstruction is assured during production.

(c) Details of the seat, disc, and stem assembly and all other replaceable internal trim.

(d) Layout of the pressure-containing envelope (body, bonnet, and bottom cap) giving dimensions which control compression of the spiral-wound gaskets. This is to assure that where remachining is necessary to repair the gasket-sealing surfaces on these parts, that compensating cuts can be accurately made to restore original gasket compression. This layout shall also specify the dimensional limits of such corrective remachining within which function of the valve remains unaffected.

(e) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.

(f) Tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).

(g) Mark areas to be radiographic, magnetic particle, or dye penetrant inspected.

3.8.2 Limited rights legend. When the Government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No. _____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

3.8.3 Technical manual. In addition to the general requirements covered by the data ordering document, the following unique technical features shall be included:

(a) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center (NAVSEC). Drawings in accordance with 3.8 and 3.8.1, shall be submitted prior to authorization of qualification tests. Qualification shall consist of the examination and tests specified in 4.3.2 through 4.3.11.

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4.3.1 Qualification inspection samples. Two valves shall be submitted for qualification examination and testing. In addition to these valves, the following additional items shall be provided:

(a) Internal replaceable trim (throttling elements, poppets, springs, pistons, etc.) - - 2 complete sets.
(b) Special tools required for installation, operation, and maintenance - - 2 complete sets.
(c) Spiral wound gaskets and piston rings - - 4 complete sets.
(d) Instruction book for installation, operation, and maintenance - - 2 complete sets.
(e) Two complete sets of drawings in accordance with 3.8 and 3.8.1.

One of the qualification inspection sample valves shall have a line tapped between the pilot and operating piston for installation of a pressure gage. This line shall terminate in a 1/8 inch NPT fitting. This valve shall also have two accelerometer blocks welded to the body at the approximate positions shown on figure 1. The accelerometer block on the body shall as far as possible be centered over the region of exit flow from the quiet-flow regulating element. Blocks shall be circular steel 1 inch in diameter and 3/8 inches thick. They shall have a surface flatness of 700 micro-inches root mean square (r.m.s.) or better and a surface finish of 125 micro-inches r.m.s. or better. Center of each block shall be drilled and tapped to a depth of at least 1/4 inch with 10-32 N.F. threads. The tapped holes shall be perpendicular to the finished surfaces within plus or minus 1 degree. The blocks shall be preserved with a rust inhibiting coating.

4.3.2 Examination prior to testing. Upon receipt of the qualification inspection samples, the samples shall be disassembled and visually and dimensionally examined to determine conformance with the requirements of this specification and the valve drawings. On reassembly, all gaskets and piston rings shall be replaced with spares randomly selected from those provided with the valves. Any difficulty encountered in assembly or disassembly shall be noted.

4.3.2.1 The maintenance actions specified in 3.4.2.1 shall be demonstrated.

4.3.3 Nondestructive tests. Nondestructive tests shall be as specified in MIL-STD-278 and in accordance with MIL-STD-271. Acceptance criteria shall be in accordance with MIL-STD-278. This shall include RT, MT/PT, pressure and visual testing.

4.3.4 Hydrostatic test. The valves shall be pressurized with water at 2175 lb/in^{2}g. There shall be no external leakage, permanent distortion, or structural failure. The diaphragm assembly may be blanked off during this test to prevent damage.

4.3.5 Dead-end test. With 740 lb/in^{2}g steam at the inlet, the valve shall be tested for seat tightness to the criteria of 3.6.3 set at 130 lb/in^{2}g. If downstream volumes or times other than specified in 3.6.3 are used, the allowable rise shall be adjusted accordingly. The downstream volume shall be free of leakage and shall be insulated to prevent condensation.

4.3.6 External leakage test. With 740 lb/in^{2}g steam at the inlet and the valve dead-ended at its highest setting, the valve shall be tested for external leakage. There shall be no external leakage which can be detected by use of a mirrored surface.

4.3.7 Functional test. The valve shall be subjected to the tests specified in 4.3.7.1 through 4.3.7.4. The extreme values of delivered pressure shall at all times be held within a range of 20 lb/in^{2}g. There shall be no evidence of hunting, chatter or any other unstable or unsatisfactory operation of the valve either continuous or transient, at any time during these tests. The valve shall be continuously monitored for noise measurements during these tests, and at all times shall meet the noise criteria of 3.6.5. In addition to the above specific testing, functional testing shall comprise a total of not less than 20 hours steam system testing.

4.3.7.1 Flow-droop test. With 250 lb/in^{2}g steam pressure maintained at the inlet, the valve shall be set at shut-off (dead-end) to deliver 130 lb/in^{2}g. Flow shall then be increased in increments of approximately 500 lb/hr until a rate of 10,000 lb/hr is reached and then decreased in the same sequence to shut-off. Without readjusting the valve, inlet pressure shall be increased to a constant level of 740 lb/in^{2}g and the above flow change sequence shall be repeated. A continuous record of delivered pressure shall be maintained. The extremes of delivered pressure during all of the above operations shall be noted and the maximum difference shall not exceed 20 in/lb^{2}g.

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4.3.7.2 The tests of 4.3.7.1 shall be repeated except with the initial delivered pressure setting at shut-off of 200 lb/in^{2}g. The same 20 lb/in^{2}g maximum deviation acceptance criteria shall apply.

4.3.7.3 Flow transient response test. With 600 lb/in^{2}g steam pressure maintained at the inlet, the valve shall be set at shut-off (dead-end) to deliver 165 lb/in^{2}g. Flow shall shall be varied between 500 and 10,000 lb/hr at a rate of approximately 800 lb/hr per second in both increasing and decreasing directions. All performance parameters shall be recorded, with particular emphasis on detection of transient noise exceeding the levels permitted in 3.6.5. The extremes of delivered pressure during all of the above operations shall not exceed 20 lb/in^{2}g.

4.3.7.4 Inlet pressure transient test. With 250 lb/in^{2}g inlet steam pressure, the valve shall be set at shut-off (dead-end) to deliver 165 lb/in^{2}g. A flow of approximately 500 lb/hr shall then be established and maintained. Inlet pressure shall be varied between 250 and 740 lb/in^{2}g at a rate of approximately 30 lb/in^{2}g per second in both increasing and decreasing directions. This test shall then be repeated at flow rates of approximately 4,000 and 10,000 lb/hr. All performance parameters shall be recorded, with particular emphasis on detection of transient noise exceeding the levels permitted in 3.6.5. The extremes of delivered pressure during all of the above operations shall not exceed 20 lb/in^{2}g.

4.3.8 High-impact shock test. The valve shall be subjected to and meet the high-impact shock tests for grade A, class I of MIL-S-901 pressurized to 740 lb/in^{2}g with water.

4.3.9 Vibration test. Valve shall be vibration tested in accordance with type I of MIL-STD-167-1 pressurized to 740 lb/in^{2}g with water. There shall be no resonance in the range of frequency tested.

4.3.10 Post test examination. After completion of the tests specified in 4.3.3 through 4.3.9, the test valve shall be disassembled and visually and dimensionally examined. Any damage, excessive wear, or signs of galling or pitting shall be cause for rejection.

4.3.11 Endurance test. The valve shall be operated for a total of 750 hours under shipboard conditions. It shall operate at all times without evidence of instability, without allowing delivered pressure to vary from specified limits, without deviating from the noise requirements of 3.6.5, and without requiring any maintenance, repair, or adjustment effort. At the successful completion of the 750 hours of operation, the valve shall be removed and completely disassembled. All parts shall be examined for signs of excessive wear or any other condition indicating impending failure or malfunction. Any such condition shall constitute grounds for failure of the valve to pass these tests, regardless of how satisfactorily the valve performed during the 750 hours of operation. Impending failure or malfunction is defined as one which can be expected to occur within one year of operation.

4.4 Quality conformance inspection. Each production valve shall be examined and tested as specified in 4.4.1 through 4.4.6.

4.4.1 Visual and dimensional examination. Each valve shall be visually and dimensionally examined to verify conformance with the requirements of this specification not requiring tests and the engineering drawings.

4.4.2 Nondestructive tests. Nondestructive tests as specified in 4.3.3 shall be conducted on each valve.

4.4.3 Hydrostatic tests. Hydrostatic tests as specified in 4.3.4 shall be conducted on each valve.

4.4.4 Dead-end test. Dead-end tests as specified in 4.3.5 shall be conducted on each valve.

4.4.5 External leakage. External leakage tests as specified 4.3.6 shall be conducted on each valve.

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4.4.6 Flow test. Each valve shall be flow tested with 250 lb/in^{2}g inlet pressure while set at 200 lb/in^{2}g. Flow capacity shall be not less than 10,000 lb/hr while maintaining delivered pressure within the limits specified in 3.6.1. Where approved by NAVSEC, production-unit flow testing may be accomplished using only the quiet-flow throttling element or using other inlet steam pressures or with water or air or any of the preceding. The necessary correlation of flow measuring methods must be demonstrated by the contractor in order to receive such approval. Where the contractor can prove that the design and method of fabrication assures that all production units meet specified flow requirements, waiver of production-unit flow testing will be considered.

4.5 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B or C and marked in accordance with MIL-V-3, as specified (see 6.2.1).

5.1.1 Special marking. In addition to the marking, as applicable, required in 5.1 and 5.1.2, shelf life marking shall be applied. Shipping containers (see 5.1) shall be stenciled in accordance with the marking instructions of MIL-STD-740.

5.1.2 Repair parts. Repair parts shall be preserved-packaged, packed and marked (see 5.1.1) in accordance with the levels of MIL-R-196 as applicable for the intended use and destination as follows:

	Levels	
	Preservation-packaging	Packing
Onboard (accompanying valves)	A	C
Stock	A	B
Immediate use	C	C

Unless otherwise specified (see 6.2.1), repair parts shall be packed separately and shipped concurrently with the valve(s).

5.2 Use of polystyrene (loose-fill) material.

5.2.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2.1) use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.2.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

5.3 Unpacking instructions. Unpacking instructions, as specified in MIL-STD-740, shall be provided.

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5.4 Technical manuals. Technical manuals, which accompany shipments that are packed level A or B, shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removeability the location of the manual shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C, as specified (see 6.2.1).

5.5 Data (see 3.8). Data shall be prepared for delivery in such a manner as to insure the required information is protected against deterioration, physical damage or loss during shipment from the supplier to the receiving activity. Packages or shipping containers shall as a minimum conform to the level C requirements of 5.1.

6. NOTES

6.1 Intended use. The quiet-design steam pressure reducing valve specified herein is intended for services in a submarine steam system.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable levels of preservation-packaging and packing required (see 5.1).
(c) Levels of preservation-packaging and packing required for repair parts (see 5.1.2).
(d) Use of polystyrene (loose-fill) material (see 5.2.1).
(e) Packing of technical manuals (see 5.4).

6.2.2 Contract data requirements. When this specification is used in a procurement invoking the provisions of Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Spec paragraph	Data requirement	Service	Applicable DID	Options
(a)	3.8.1	Drawings	SH	UDI-E-23174	Categories A, G and H, form 2, types II and III
(b)	3.8.3 and 5.4	Technical manual	SH	DI-M-2043	Type I of MIL-M-15071

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24517 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

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6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4820-N305)

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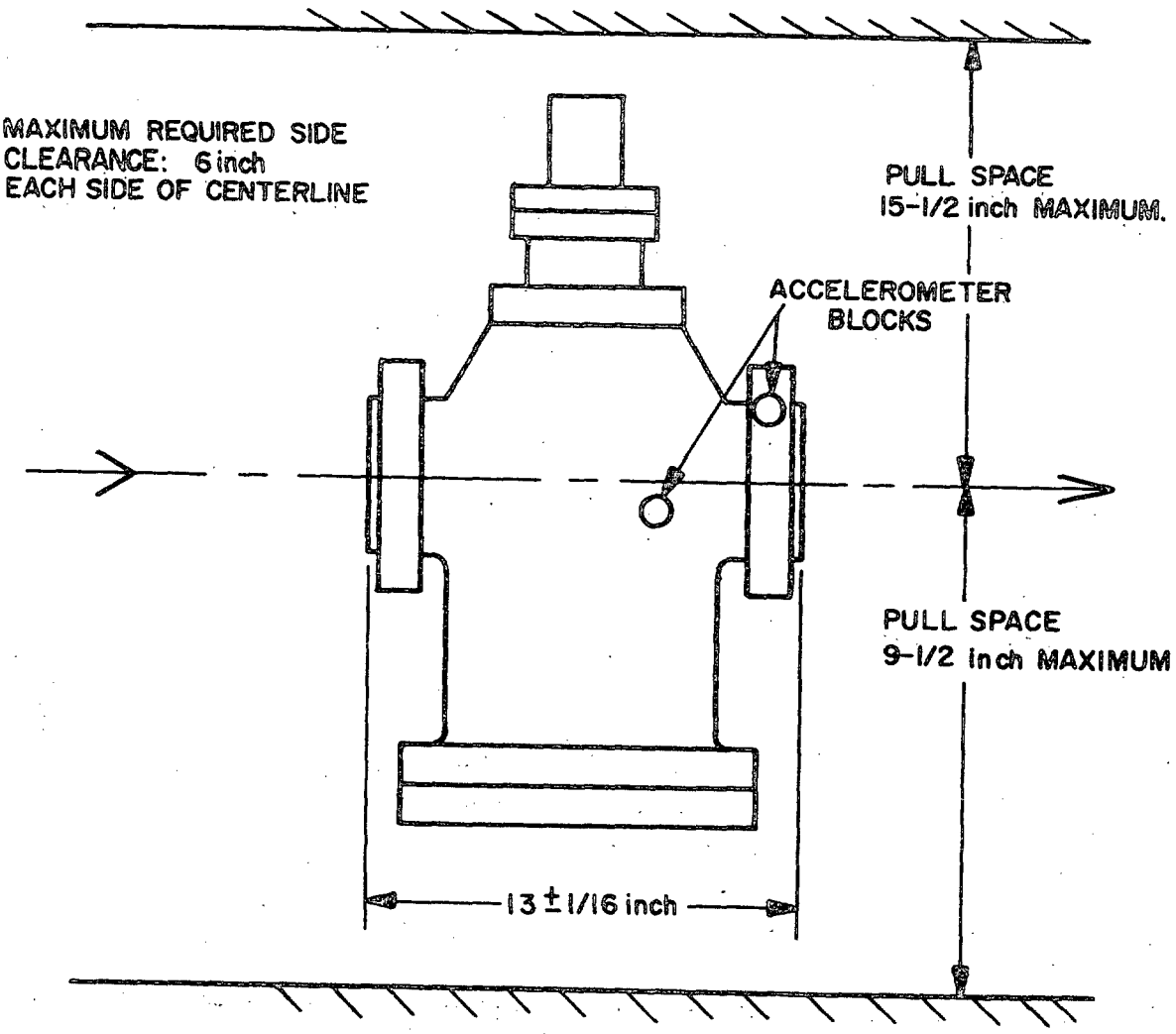


Figure 1 - Interface dimensions.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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